KAIZEN MATHS

Lesson starter

Normal Distribution and Standardisation

Inverse & Parameters

Name _____

S1

X ~ N(100, 36). Find a such that P(X < a) = 0.1500.

S2

X ~ N(μ, 9) and P(X < 64.9) = 0.9495. Find μ.

S3

X ~ N(300, σ^2) and P(X > 311.8) = 0.0250. Find σ.

S4

X ~ N(μ, σ^2), P(X < 34.9) = 0.1003, and P(X > 43.4) = 0.2005. Find μ and σ.

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STA-9E35-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Normal Distribution and Standardisation

Inverse & Parameters

Name _____

E1

$X \sim N(150, 16)$. Find a such that $P(X < a) = 0.1000$.

E2

$X \sim N(\mu, 100)$ and $P(X < 291.6) = 0.2005$. Find μ.

E3

$X \sim N(200, \sigma^2)$ and $P(X > 213.2) = 0.1357$. Find σ.

KAIZEN MATHS

Exit ticket

Normal Distribution and Standardisation

Inverse & Parameters

Name _____

E1

X ~ N(150, 16). Find a such that P(X < a) = 0.1000.

E2

X ~ N(μ, 100) and P(X < 291.6) = 0.2005. Find μ.

E3

X ~ N(200, σ²) and P(X > 213.2) = 0.1357. Find σ.

STA-9E35-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Normal Distribution and Standardisation

Inverse & Parameters

Name _____

E1

X ~ N(150, 16). Find a such that P(X < a) = 0.1000.

E2

X ~ N(μ, 100) and P(X < 291.6) = 0.2005. Find μ.

E3

X ~ N(200, σ²) and P(X > 213.2) = 0.1357. Find σ.

STA-9E35-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Normal Distribution and Standardisation

Inverse & Parameters

Name _____

E1

X ~ N(150, 16). Find a such that P(X < a) = 0.1000.

E2

X ~ N(μ, 100) and P(X < 291.6) = 0.2005. Find μ.

E3

X ~ N(200, σ²) and P(X > 213.2) = 0.1357. Find σ.

STA-9E35-L3 · E

Show your working

Teacher answers & working

Normal Distribution and Standardisation · Inverse & Parameters

STA-9E35-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

X ~ N(100, 36). Find a such that P(X < a) = 0.1500.

Answer:

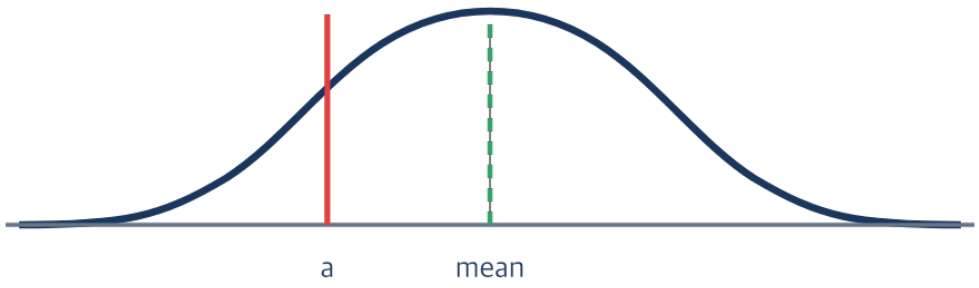
$a = 93.8$

Working:

Convert the left-tail probability into a z-value.

P(X < a) = 0.1500 means P(Z < z) = 0.1500.

$z = \Phi^{-1}(0.1500) \approx -1.04$.



Convert back from z to x.

$x = \mu + z\sigma$.

$a = 100 + (-1.04)(6) = 93.8$.

$a = 93.8$

S2

X ~ N(μ, 9) and P(X < 64.9) = 0.9495. Find μ.

Answer:

$\mu = 60$

Working:

Convert the probability to a z-value.

$P(X < 64.9) = 0.9495$, so $z = \Phi^{-1}(0.9495) \approx 1.64$.

Use the standardisation equation.

$z = (x - \mu) / \sigma$.

$1.64 = (64.9 - \mu) / 3$.

$64.9 - \mu = (1.64)(3)$.

$\mu = 64.9 - (1.64)(3) = 60$.

$\mu = 60$

Teacher answers & working

Normal Distribution and Standardisation · Inverse & Parameters

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S3

$X \sim N(300, \sigma^2)$ and $P(X > 311.8) = 0.0250$. Find σ.

Answer:

$\sigma = 6$

Working:

Change the right-tail probability into a left-tail probability.

$P(X > 311.8) = 0.0250$, so $P(X < 311.8) = 1 - 0.0250 = 0.9750$.

$z = \Phi^{-1}(0.9750) \approx 1.96$.

Use the standardisation equation.

$1.96 = (311.8 - 300) / \sigma$.

$1.96\sigma = 11.8$.

$\sigma = 11.8 / 1.96 = 6$.

$\sigma = 6$

S4

$X \sim N(\mu, \sigma^2)$, $P(X < 34.9) = 0.1003$, and $P(X > 43.4) = 0.2005$. Find μ and σ.

Answer:

$\mu = 40, \sigma = 4$

Working:

Convert each probability into a z-value.

$P(X < 34.9) = 0.1003$, so $z_1 = \Phi^{-1}(0.1003) \approx -1.28$.

$P(X > 43.4) = 0.2005$, so $P(X < 43.4) = 0.7995$ and $z_2 \approx 0.84$.

Write two equations using $x = \mu + z\sigma$.

$34.9 = \mu + (-1.28)\sigma$.

$43.4 = \mu + (0.84)\sigma$.

Subtract the equations to find σ.

$43.4 - 34.9 = (0.84 - (-1.28))\sigma$.

$\sigma = 8.5 / 2.12 = 4$.

Substitute back to find μ.

$\mu = 34.9 - (-1.28)(4) = 40$.

$\mu = 40, \sigma = 4$

Teacher answers & working

Normal Distribution and Standardisation · Inverse & Parameters

STA-9E35-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

X ~ N(150, 16). Find a such that P(X < a) = 0.1000.

Answer:

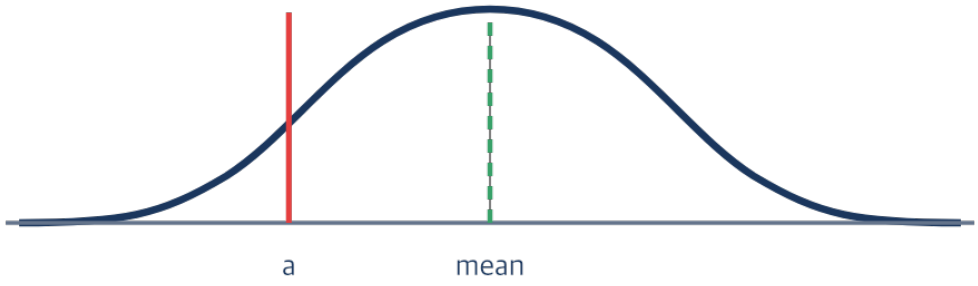
$a = 144.9$

Working:

Convert the left-tail probability into a z-value.

P(X < a) = 0.1000 means P(Z < z) = 0.1000.

$z = \Phi^{-1}(0.1000) \approx -1.28$.



Convert back from z to x.

$x = \mu + z\sigma$.

$a = 150 + (-1.28)(4) = 144.9$.

$a = 144.9$

E2

X ~ N(μ, 100) and P(X < 291.6) = 0.2005. Find μ.

Answer:

$\mu = 300$

Working:

Convert the probability to a z-value.

$P(X < 291.6) = 0.2005$, so $z = \Phi^{-1}(0.2005) \approx -0.84$.

Use the standardisation equation.

$z = (x - \mu) / \sigma$.

$-0.84 = (291.6 - \mu) / 10$.

$291.6 - \mu = (-0.84)(10)$.

$\mu = 291.6 - (-0.84)(10) = 300$.

$\mu = 300$

Teacher answers & working

Normal Distribution and Standardisation · Inverse & Parameters

STA-9E35-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

X ~ N(200, σ^2) and P(X > 213.2) = 0.1357. Find σ.

Answer:

$\sigma = 12$

Working:

Change the right-tail probability into a left-tail probability.

$P(X > 213.2) = 0.1357$, so $P(X < 213.2) = 1 - 0.1357 = 0.8643$.

$z = \Phi^{-1}(0.8643) \approx 1.10$.

Use the standardisation equation.

$1.10 = (213.2 - 200) / \sigma$.

$1.10\sigma = 13.2$.

$\sigma = 13.2 / 1.10 = 12$.

$\sigma = 12$